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FIREARM WARNING ASSEMBLY

Abstract

A firearm warning assembly includes a band comprising a first edge region having an insert hole and a second edge region opposite the first edge region having an attachment hole and an attachment insert. The attachment insert includes a base and a post extending from the base and engageable with the insert hole. A method includes engaging an attachment insert comprising a base and a post extending from the base with an insert hole in a first edge region of a band, positioning the band around a firearm, and engaging the post with an attachment hole in a second edge region of the band opposite the first edge region to secure the band to the firearm.

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Background/Summary

BACKGROUND

[0001] The present disclosure relates generally to a warning system for a firearm, such as a firearm

warning assembly including a band that engages the firearm and an indicator that indicates an unloaded state of the firearm.
[0002] Various devices may be used to increase firearm safety. Lock boxes or safes may restrict access to a firearm. Bands or locks may cover or inhibit operation of a trigger mechanism for discharging a firearm. Such devices do not provide an indication of the loaded or unloaded state of the firearm.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0003] The present disclosure may be better understood, and its numerous features and advantages made apparent to those skilled in the art, by referencing the accompanying drawings. The use of the same reference symbols in different drawings indicates similar or identical items.
[0004] FIGS. **1**A and **1**B are front isometric views of a firearm warning assembly, in accordance with some embodiments.
[0005] FIGS. **2**A and **2**B are back isometric views of the firearm warning assembly, in accordance with some embodiments.
[0006] FIG. **3** is an isometric view of an attachment insert, in accordance with some embodiments.
[0007] FIGS. **4**A and **4**B are front are isometric views of chamber plug inserts, in accordance with some embodiments.
[0008] FIGS. **5**A and **5**B are isometric views of the firearm warning assembly being installed on a firearm, in accordance with some embodiments.
DETAILED DESCRIPTION
[0009] FIGS. **1**A, **1**B, **2**A, **2**B, **3**, **4**A, **4**B and **5** illustrate example embodiments of a firearm warning assembly **100**. FIGS. **1**A and **1**B are front isometric views of the firearm warning assembly **100** and FIGS. **2**A and **2**B are back isometric views of the firearm warning assembly **100**, in accordance with some embodiments. In some embodiments, the firearm warning assembly **100** comprises a band **102**, an attachment insert **104**, and a chamber plug insert **106**. FIG. **3** is an isometric view of the attachment insert **104**, in accordance with some embodiments. FIGS. **4**A and **4**B are front are isometric views of chamber plug inserts **106**, in accordance with some embodiments. The front views of FIGS. **1**A and **1**B represent the outer surface **102**S**2** of the firearm warning assembly **100** that would be visible when the firearm warning assembly **100** is installed on a firearm, and the back views of FIGS. **2**A and **2**B represent the inner surface **102**S**1** of the firearm warning assembly **100** that would engage the firearm.
[0010] In some embodiments, the band **102** includes insert holes **102**A to facilitate positioning of the attachment insert **104** and the chamber plug insert **106** and attachment holes **102**B for engaging the attachment insert **104**. The holes **102**A, **102**B may be arranged in rows. The number of rows and the number of holes **102**A, **102**B in each row may vary to provide flexibility for positioning the inserts **104**, **106**. Recesses **102**R may be provided proximate the insert holes **102**A to allow the inserts **104**, **106** to be inset with respect to an inner surface **102**S**1** of the band **102**. The position of the attachment insert **104** in the insert holes **102**A may vary depending on the size of the firearm such that the band **102** is stretched when installed on the firearm to increase the reliability of the engagement. The band **102** may be expandable and contractible along more than one axis. In some embodiments, the attachment insert **104** may be inserted into the holes **102**B and engage the holes **102**A to secure the band **102**. In some embodiments, the band **102** comprises an indicia window **102**W where warning text or instructions may be provided and visible to the user when the firearm warning assembly **100** is installed on a firearm.
[0011] In some embodiments, the chamber plug insert **106** engages a chamber of the firearm to provide an indication that the chamber is empty and the firearm is unloaded. If the firearm were to

be loaded, the chamber plug insert **106** would not interface with the firearm. Thus, the chamber plug insert **106** provides a visual indication of the unloaded state of the firearm. The position of the chamber plug insert **106** in the insert holes **102**A may vary depending on the relative position of the chamber of the firearm.

[0012] In some embodiments, as shown in more detail in FIG. **3**, the attachment insert **104** comprises a base **104**B and posts **104**H, **104**P extending from the base **104**B. In the illustrated embodiment, the posts **104**H have hook shaped ends, and the post **104**B has a ball shaped end. The number of posts **104**H, **104**P may vary. Flanges **104**F may be provided on the posts **104**H, **104**P in intermediate positions along the length of the posts **104**H, **104**P for engaging the band **102**, such that the band **102** is retained between the base **104**B and the flanges **104**F.

[0013] In some embodiments, as shown in more detail in FIGS. **4**A and **4**B, the chamber plug insert **106** comprises a base **106**B, posts **106**F extending from the base **106**B, a chamber plug **106**P, and an indicator **1061** extending from the base **106**B. In some embodiments, the indicator **1061** is positioned in a middle region of the base **106**B and the chamber plug **106**P is positioned in an end region and axially aligned with one of the posts **106**F. Alternatively, the indicator **1061** and the chamber plug **106**P may be axially aligned. The number of posts **106**F may vary. In the illustrated embodiment, the posts **106**F have flanged ends for engaging the band **102**, such that the band **102** is retained between the base **106**B and the posts **106**F. The indicator **1061** extends from the base **106**B in a direction opposite to the direction of the chamber plug **106**P such that the indicator **1061** is visible on an outside surface **102**S**2** of the band **102** when installed on a firearm to provide a visual indication of the unloaded state of the firearm. The chamber plug **106**P has an offset portion **1060** extending from the base **106**B into the receiver of the firearm such that a plug base portion **106**S positions the plug **106**P such that it passes through the center of the firearm. A base portion **106**S of chamber plug **106**P is sized to approximately ride on the inside chamber wall of the firearm. The chamber plug **106**P extends into the breech of the chamber of the firearm when the firearm is in an unloaded state.

[0014] In the embodiment of FIG. **4**A, the chamber plug **106**P is elongated and curved to facilitate insertion of the chamber plug **106**P into the chamber of a shotgun or a rifle with a relatively long breech. FIG. **4**B illustrates an alternative embodiment of the chamber plug **106**P that is barrel shaped to facilitate insertion of the chamber plug **106**P into the chamber of a handgun with a shorter breech.

[0015] FIGS. **5**A and **5**B are isometric views of a firearm warning assembly **100** being installed on a firearm **108**, in accordance with some embodiments. When the firearm warning assembly **100** is installed on the firearm **108**, the band **102** is wrapped around the firearm **108** to prevent access to the chamber. In some embodiments, the band **102** may also be wrapped around the trigger guard if the chamber and trigger guard are aligned, for example, with a handgun. A user may select the chamber plug insert **106** appropriate to the firearm **108** (e.g., the chamber plug insert **106** of FIG. **4**A or FIG. **4**B) and position the chamber plug insert **106** in the insert holes **102**A depending on the location of the chamber. The user inserts the chamber plug **106**P of the chamber plug insert **106** into the chamber of the firearm as show in in FIG. **5**A and then wraps the band **102** around the firearm **108**. The user applies enough force to at least slightly stretch the band **102** to align the posts **104**H, **104**P with the insert holes **102**A in the band **102** and engage the band **102** as shown in FIG. **5**B. The stretching of the band **102** allows better engagement with the posts **104**H with hook shaped ends. The indicator **1061** extends is visible on an outside surface **102**S**2** of the band **102** when installed on the firearm **108** to provide a visual indication of the unloaded state of the firearm **108**.

[0016] Note that not all of the activities or elements described above in the general description are required, that a portion of a specific activity or device may not be required, and that one or more further activities may be performed, or elements included, in addition to those described. Still further, the order in which activities are listed are not necessarily the order in which they are performed. Also, the concepts have been described with reference to specific embodiments. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the present disclosure as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present disclosure.

[0017] Benefits, other advantages, and solutions to problems have been described above with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any feature(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential feature of any or all the claims. Moreover, the particular embodiments disclosed above are illustrative only, as the disclosed subject matter may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. No limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular embodiments disclosed above may be altered or modified and all such variations are considered within the scope of the disclosed subject matter. Accordingly, the protection sought herein is as set forth in the claims below.

Claims

1. A firearm warning assembly, comprising: a band comprising a first edge region having an insert hole and a second edge region opposite the first edge region having an attachment hole; and an attachment insert, comprising: a base; and a post extending from the base and engageable with the insert hole.

2. The firearm warning assembly of claim 1, wherein: the post has a hook shaped end portion.

3. The firearm warning assembly of claim 1, wherein: the band comprises a second insert hole in the first edge region and a second attachment hole in the second edge region; the attachment insert comprises a second post extending from the base and engageable with the second insert hole; the post comprises a hook shaped end portion; and the second post comprises a ball shaped end portion.

4. The firearm warning assembly of claim 1, wherein: the band comprises a recess; and the base of the attachment insert is positionable in the recess.

5. The firearm warning assembly of claim 1, wherein: the post of the attachment insert has a flange positioned in an intermediate position along a length of the post.

6. The firearm warning assembly of claim 1, wherein: the band comprises rows of holes in the first edge region; and the insert hole is in one of the rows of holes.

7. The firearm warning assembly of claim 1, comprising: a chamber plug insert engageable with the band and comprising: a second base; a chamber plug extending from the second base in a first direction; and an indicator extending from the second base in a second direction opposite the first direction.

8. The firearm warning assembly of claim 7, wherein: the band comprises a second insert hole in one of the first edge region or the second edge region; and the chamber plug insert comprises a second post extending from the second base in the second direction and engageable with the second insert hole.

9. The firearm warning assembly of claim 8, wherein: the band comprises rows of holes in the first edge region; the insert hole is in a first row of the rows of holes; and the second insert hole is in a second row of the rows of holes.

10. The firearm warning assembly of claim 8, wherein: the second post has a flange shaped end portion.

11. A method, comprising: engaging an attachment insert comprising a base and a post extending from the base with an insert hole in a first edge region of a band; positioning the band around a firearm; and engaging the post with an attachment hole in a second edge region of the band opposite the first edge region to secure the band to the firearm.

12. The method of claim 11, wherein: the post of the attachment insert has a hook shaped end portion; and engaging the post with the attachment hole comprises stretching the band to engage the post with the attachment hole.

13. The method of claim 11, wherein: the band comprises a recess; and engaging the attachment insert comprises positioning the base of the attachment insert in the recess.

14. The method of claim 11, wherein: the post of the attachment insert has a flange positioned in an intermediate position along a length of the post; and engaging the attachment insert comprises positioning the band between the base and the flange.

15. The method of claim 11, wherein: the band comprises rows of holes in the first edge region; and engaging the attachment insert comprises selecting the insert hole in one of the rows of holes for positioning the attachment insert.

16. The method of claim 11, comprising: engaging a chamber plug insert comprising a second base, a chamber plug extending from the second base in a first direction, and an indicator extending from the second base in a second direction opposite the first direction with the band; and inserting the chamber plug in a chamber of the firearm prior to engaging the post with the attachment hole to secure the band to the firearm.

17. The method of claim 16, wherein: the band comprises a second insert hole in one of the first edge region or the second edge region; the chamber plug insert comprises a second post extending from the second base in the second direction; and engaging the chamber plug insert comprises engaging the second post with the second insert hole.

18. The method of claim 17, wherein: the band comprises rows of holes in the first edge region; the insert hole for engaging the attachment insert is in a first row of the rows of holes; and engaging the chamber plug insert comprises selecting the second insert hole in a second row of the holes different than the first row.

19. An apparatus, comprising: a firearm comprising a chamber and a control; a firearm warning assembly, comprising: a band comprising a first edge region having an insert hole and a second edge region opposite the first edge region having an attachment hole; and an attachment insert, comprising: a base; and a post extending from the base and engageable with the insert hole and the attachment hole to position the band over the control of the firearm; a chamber plug insert engageable with the band and comprising: a second base; a chamber plug extending from the second base in a first direction and positioned in the chamber of the firearm; and an indicator extending from the second base in a second direction opposite the first direction.

20. The apparatus of claim 19, wherein: the band comprises a second insert hole in one of the first edge region or the second edge region; and the chamber plug insert comprises a second post extending from the second base in the second direction and engageable with the second insert hole.